

Question	Answer	Marks
3(a)(i)	$F = W / d$ $= 5.1 \times 10^3 / 23$	C1
	= 220 N	A1

Question	Answer	Marks
3(a)(ii)	$(\Delta)E_{(P)} = (16.1 \times 10^3 - 1.3 \times 10^3 - 5.1 \times 10^3)$ $= 9.7 \times 10^3$	C1
	$(\Delta)E_{(P)} = mg(\Delta)h$	C1
	$\Delta h = (9.7 \times 10^3) / (280 \times 9.81)$ $= 3.5 \text{ m}$	A1
3(b)	kinetic energy (at Z) is less (than at X) (due to work done by carriage against resistive forces), so the speed is less (at Z than at X)	B1